

Market Unfairness: a Model-Free Approach to Cryptocurrency Option Market Quality

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Abstract

Put–call parity (PCP) ties a European call, a put, and a matching future into a one-to-one correspondence, so their *standing quotes* on a single venue should be mutually consistent up to transaction costs. We propose *market unfairness* (MU): the residual PCP violation in top-of-book quotes after netting out direct transaction costs. Unlike the PCP literature, which tests trade prices, MU is a statement about quotes—a model-free, cost-robust symptom of imperfect liquidity provision, in the spirit of the best-execution rules that police quote quality for instruments that should trade at parity. It does not fully characterize a market’s liquidity, but it provides a clean, model-free benchmark for comparison across venues, products, and time.

We apply this framework to high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching futures over 2020–2026. We characterize the level and cross-section of MU across exchanges, underlyings, moneyness, and maturity, and document its evolution through the 2024 regulatory shift (U.S. spot Bitcoin and Ether ETF approvals and MiCA). MU differs only moderately across venues and underlyings but varies significantly over the option surface and compressed markedly around 2024. Regression analysis confirms that the compression is not fully explained by the standard liquidity frictions (spreads, depth, staleness, Amihud ratio, spread recovery time). The residual improvement is instead consistent with the entry of participants whose quoting behavior enforces put–call parity more tightly—an improvement in quote *quality* rather than a mere increase in depth or a tightening of spreads.

Keywords: cryptocurrency options, bid–ask spreads, market unfairness, put–call parity, limits to arbitrage, ETF approval, MiCA

JEL classification: G12, G14, G15, G23

Contents

1	Introduction	3
2	Related literature	5
2.1	The limits of put-call-parity arbitrage in general	5
2.2	PCP in crypto derivative markets	5

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3	Liquidity and frictions in the crypto option market	6
4	Market Unfairness: concept and measurement	9
4.1	Market maker advantage and market unfairness precisely defined	9
4.2	Description of arbitrage configurations	11
4.3	The no-MU band and the shape of the MMA distribution	12
4.4	Data	13
4.5	Panel construction	14
5	Empirical structure of the MU	16
5.1	The shape of the MMA distribution	16
5.2	U_u across exchanges, moneyness, and maturity	16
6	The 2024 regulatory shift	20
6.1	MU over time	20
7	Regression analysis	23
7.1	Empirical design	23
7.2	Results	23
8	Robustness	26
9	Conclusion	28

1 Introduction

This paper studies the top-of-book quotes of cryptocurrency option markets, and the premium that market makers require to quote them. The object is not transaction prices, but standing quotes: the quality of the top of as a measure the market’s efficiency. For example, the smaller the bid-ask spread, the better the quality of the market, the closer the market conforms to the law of one price (Demsetz 1968, Amihud & Mendelson 1986).

There is no general theory of how wide a spread *should* be, what *would* be fair, but it is understood that it is compensation for the market maker to provide liquidity (Demsetz 1968, Ho & Stoll 1981, Glosten & Milgrom 1985). How much compensation is fair is a matter of what model we choose for the behavior of market makers (Ho & Stoll 1981, Bollen et al. 2004). However, there are some special cases where we *can* identify clearly unfair quotes. The best known is the case where identical instruments are quoted on several venues with no friction between transacting on one or another: this is the Nasdaq. There the best-execution rules under Regulation NMS (U.S. Securities and Exchange Commission 2005) forbid filling retail market order flow at a price worse than the National Best Bid and Offer across venues.

There are other situations which might seem similar: The ETF and ADR markets, as well as the On/Off-the-run Treasury markets (Pan & Zeng 2017, Gagnon & Karolyi 2010, Krishnamurthy 2002). But here there are real frictions (liquidity and market access) that legitimately keep the ETF market price vs ETF creation/redemption cost, or ADR and equity price, or on/off-the-run Treasury price, from being equal.

In this paper we show a situation where there are equivalent instruments traded on a single market, where price differentials should not exist at all: These are markets in European options with matching futures. The put-call parity (PCP) relationship establishes a one-to-one correspondence between a put, a futures, and a call. The breaking of the PCP has been extensively studied in various markets (Klemkosky & Resnick 1979, Kamara & Miller 1995, Cremers & Weinbaum 2010, Ofek et al. 2004), and it has been shown that most apparent PCP breaking is due to transaction costs involved in replicating a call from a put. PCP breaking remaining beyond what can be explained by transactions costs are assumed to be idiosyncratic.

We propose to use the PCP breaking remaining in quotes after accounting for transaction costs for a new purpose: a measure of the quality of a market or venue for particular option chains. This measure we propose to call market unfairness (MU), where “unfairness” is a standardized statistic rather than a welfare judgment: it flags quotes that sit outside the cost-implied no-arbitrage band, a symptom of imperfect quote consistency. Market unfairness is not meant to characterize a venue’s market quality completely, but it does provide a model-free measure for comparison as long as trading costs are known. We will show that MU comparisons among regimes, venues or products are robust to the cost parameter.

Our paper will focus on characterizing MU on cryptocurrency option markets. We describe empirical facts about market unfairness, including its response to regulatory changes in 2024. We find evidence corroborating the fact that the regulatory changes have introduced new players whose market making behavior improves quoting quality, not just increased volumes or tightened spreads.

Section 2 places market unfairness in the literature. Section 3 describes the market through the standard liquidity measures. Section 4 defines market maker advantage MMA, MU, and describes the data and panel. Section 5 presents the level and cross-section of

market unfairness. Section 6 describes the 2024 regulatory shift. Section 6.1 documents the regulation induced compression and its robustness to assumptions. Section 7 reports regressions . Section 9 concludes.

2 Related literature

2.1 The limits of put-call-parity arbitrage in general

First, there is plenty of literature linking PCP breaking to bid-ask spreads. Amihud & Mendelson (1986) show spreads affect prices because investors require compensation for illiquidity. Shleifer & Vishny (1997) stress that textbook arbitrage is neither riskless nor free in practice. Testing put-call parity has a long tradition in equity, index, and currency options. Early work shows apparent violations are mostly explained by bid-ask spreads, execution timing, short-sale constraints, taxes, and trading costs (Klemkosky & Resnick 1979, Nisbet 1992, Kamara & Miller 1995, El-Mekkaoui & Flood 1998, Cremers & Weinbaum 2010).

PCP arbitrage involves opening a costless portfolio of put, call and underlying (or futures) contracts which has strictly positive payout. This can be phrased as a real and synthetic price for an option, which are supposedly equal. The split of the real-synthetic price difference into direct costs plus a residual is customary, and the study of this residual is the object of the PCP breaking literature. Market unfairness is the analogous residual for the parity quote, the object of this paper. Pontiff (1996, 2006) shows holding costs—including the idiosyncratic risk of an undiversified arbitrageur—dominate, but mispricing survives. Chen et al. (2021) estimate the no-arbitrage band directly from data. Gromb & Vayanos (2010) survey the theory: frictions that make convergence risky widen the band. For the market maker, Bollen et al. (2004) split the quoted spread into order processing, inventory holding, and adverse selection.

2.2 PCP in crypto derivative markets

Crypto options add complications: contracts are often inverse or quanto, collateral and settlement currencies differ from payoff currencies, and the reference index is not always tradable (Alexander et al. 2023, Lucic & Sepp 2024). Gârleanu & Pedersen (2011) show how margin and funding constraints create deviations from the law of one price. Alexander et al. (2024) link Bitcoin and Ether options to perpetual futures and find efficiency improving over time, especially at long maturities. Felföldi-Szűcs et al. (2024) study high-frequency parity in Binance Ether options and show violations shrink once synchronization and costs are handled. We extend this to inverse options and matching futures on Deribit and OKX.

3 Liquidity and frictions in the crypto option market

Direct quote frictions over time by exchange and underlying (1-month rolling mean, over strike and maturity)

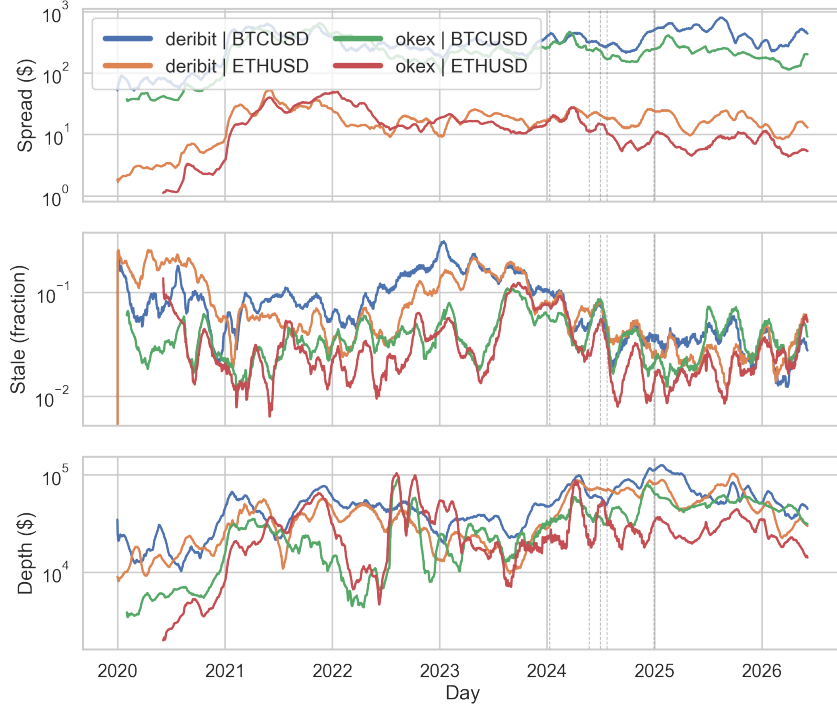


Figure 1: Raw quote statistics (aka frictions) over time by exchange and underlying, smoothed with a trailing 1-month rolling mean and aggregated over strike and maturity. Bid–ask spread in dollars ($\text{Spr}_{g,t}^{\$}$), stale-quote fraction $\text{Stale}_{g,t}$, and quote depth in dollars. The spread and depth dollar values are measured for one trade unit. Dashed vertical lines mark the 2024 events: BTC ETP approval (10 Jan), ETH ETF approval (23 May), MiCA stablecoin rules (30 Jun), ETH ETF trading (23 Jul), and MiCA CASP rules (30 Dec).

Before proposing a new measure, we describe the crypto option market through the classical liquidity metrics. For each exchange and underlying we track several measures over 2020–2026, aggregating over strike and maturity to a daily series and smoothing with a one-month rolling mean (Figures 1 and 2).

Some are direct properties of the standing quotes:

- the average put–call bid–ask *spread* in units of equivalent dollar amount of the price difference $\text{Spr}^{\$}$
- the quote *depth* $\text{Depth}^{\$}$, the dollar-equivalent size resting at the top of book. Note that this is sensitive to the contract size convention, which we took into account.
- the *staleness* Stale , the fraction of quotes that fail to refresh within the sampling interval. Note that this staleness does not necessarily mean a market making failure, this can occur when the market simply does not move.

We also consider the following dynamic illiquidity and resiliency measures, computed for each strike–maturity cell from its intraday series and averaged over the strike–maturity surface:

Dynamic frictions over time by exchange and underlying (1-month rolling mean, over strike and maturity)

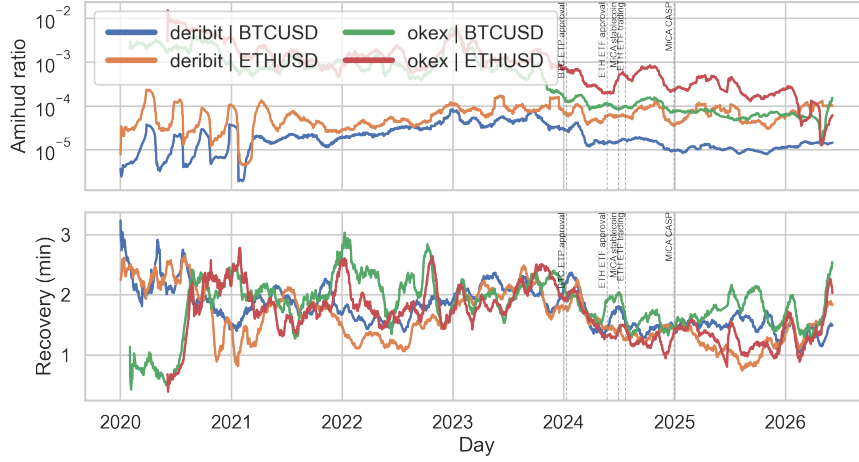


Figure 2: Dynamic measures over time by exchange and underlying, smoothed with a trailing 1-month rolling mean; per cell-day from the intraday five-minute series and averaged over the surface. The Amihud illiquidity ratio (using premium volume), and the dollar-spread recovery time. This is the AR(1) decay half-life of the dollar spread's deviation from a rolling one-hour baseline, fit on windows after a large 2σ widening, in minutes, median across the surface.

Daily premium volume by exchange and underlying (1-month rolling mean)

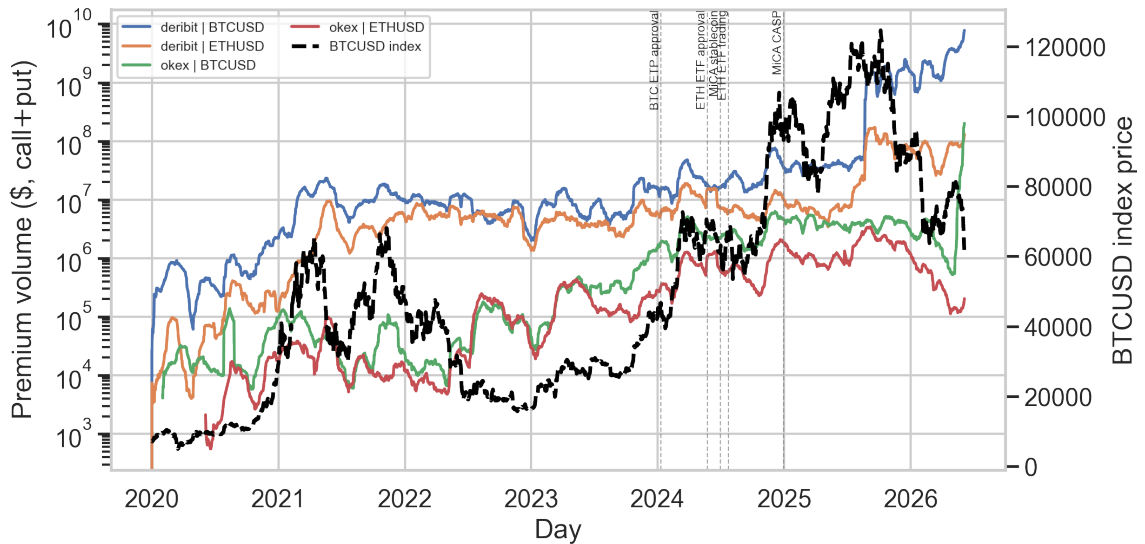


Figure 3: Daily premium volume by exchange and underlying—the traded call and put premium in dollars (traded size times the option price times the index), summed over the strike–maturity surface—1-month rolling mean, log scale, with the BTCUSD index overlaid on the right axis.

- the *Amihud* illiquidity ratio (Amihud 2002), the average of $|\text{return}|/\text{volume}$ over the cell’s five-minute call and put series—how far the price moves per unit traded—with volume measured in option-*premium* dollars (contracts times contract size times the sum of the call and put mid prices);
- the *spread recovery time*, a resiliency measure—the third dimension of liquidity alongside tightness and depth (Kyle 1985, Foucault et al. 2013). After a *large* widening—the spread’s deviation from its centered rolling one-hour baseline crossing above two robust standard deviations (from the day’s median absolute deviation)—we fit an AR(1) decay to the deviation over the following hour and report its half-life $-\ln 2 / \ln \rho$ in minutes, the time for a shock to die out by half; we compute it for both the bp and the dollar spread. Conditioning on genuine widenings, in the spirit of the order-book resiliency literature (Degryse et al. 2005, Large 2007), discards the small idiosyncratic quote wiggles that otherwise leave an unconditional AR(1) near unit-root and uninformative.

An interesting opportunity to observe the market’s liquidity evolution is the 2024 regulatory and institutional regime shift (Section 6). Figure 1 staleness falling markedly around 2024. Dollar depth rises continuously, showing a general deepening of the market over time, whereas the dollar spread shows no trend. The Amihud illiquidity ratio and the spread-recovery times (Figure 2) also show regime change. Traded volume (Figure 3) is also on a seemingly unstoppable exponential growth path, which does not obviously display a regime change in 2024.

Each channel captures a facet of liquidity, each of them are on a flat or improving trend, but can we say that an investor is better off now than before? Large players can be happy as the Amihud and recovery statistics (which characterize market impact related loss) have definitely improved, but retail investors, who in general will care about market impact, will on average they face non-improving spreads.

The rest of this paper develops MU, a complementary, model-free measure to test market fairness that is orthogonal to the above market liquidity measures. It is built only from standing quote information and applies PCP consistency conditions to filter out unfair markets. We’ll also show that this measure decreased in 2024, showing that the regulatory changes brought benefits to the retail investor too.

4 Market Unfairness: concept and measurement

Market makers post passive orders to the book. If and when the passive order is hit, the resulting exposure can be fully hedged via a PCP arbitrage loop. For example if our bid for a call is lifted, we are long the call, which we can hedge by going short futures and a corresponding put. If we got the call too cheap, we can lock in risk-free profit. A market maker seeking risk-free profits might wait for such parity breaking to occur, and join or improve a book, waiting for takers. The expected payoff is strictly positive, even though issues of queue position and the disappearance of the arbitrage path exist. We deal with these microstructure issues by studying intraday snapshot data, where the quotes involved in the arbitrage are all present at the same time. In addition, the microstructure issues only limit the dollar value and not the sign of this opportunity.

We quantify this opportunity for market makers as the *market maker advantage* (MMA) which is the payout of being filled and fully hedged via PCP. This is negative most of the time. But there are times where the market maker can expect a profit if filled: we call the extent to which such opportunities exist *market unfairness*, MU. The occurrence of cost-adjusted parity deviations at the top of the book can reasonably be called unfair, since a market maker could improve the PCP violating quote with a strictly positive expected payoff.

4.1 Market maker advantage and market unfairness precisely defined

A source of complexity in executing our study is the fact that the most liquid contracts are inverse. We use BTCUSD and ETHUSD options on Deribit and OKX, which are quoted against a USD index, but collateral, premium, and settlement are in coin. Parity must therefore be written after converting premia and the futures leg to a common currency (Alexander et al. 2023, Lucic & Sepp 2024): the coin-denominated premia are then scaled by the conversion price S as of the time of posting the quote.

For a matched strike–expiry put-call pair the profit depends on how the arbitrage is run: backward or forward, joining the call or the put side. By convention *forward* means arbitrage involving going *long* futures and vice versa. The index p for *path* can take 4 values. In every path the maker *joins* a standing quote rather than crossing it: he posts and profits only if a taker hits the standing quote. For instance, in the backward/join-call-bid path she buys the call at C_{bid} ; in the forward/join-put-bid path she sells it at C_{bid} .

The profit formulas for the four paths (Table 1), are net of the round-trip cost of executing one *trade unit*. A trade unit is the number of contracts having notional equal to the index value. So for example on OKX, 100 contracts of BTCUSD options would correspond to a trade unit, given that the contract multiplier is 0.01 on this market.

Let $m_{i,p,t}$ be the signed dollar one-trade-unit payoff of path p for *pair* i at time t , after conversion and cost adjustment. We call each (pair, path) observation a *tickpath*: every synchronized quote (*tick*) for a matched pair gives four tickpaths.

A *tick* is indexed by (pair, time). A *pair* is indexed by (exchange, underlying, strike, expiry).

We aggregate tickpaths in *cells* (g, t) with cell g corresponding to a group of pairs with exchange, underlying, time, strike, expiration, each in some range. A cell with $n_{g,t}$ pairs has $4n_{g,t}$ tickpaths.

Market maker advantage (MMA) is the mean of $m_{i,p,t}$ taken over *all* tickpaths in a cell

Table 1: Arbitrage-profit formulas by path and contract style, in dollars per contract (one trading unit of the underlying). Each expression is net of the round-trip cost for a trade unit cost = `cost_per_notional` $\times$ `index_level` from Table 3, a proportional-to-notional cost. For inverse contracts, S_{bid} and S_{ask} are the prices at which the coin payoff is converted to USD or stablecoin.

Path	Linear contracts	Inverse contracts
fwd, join put bid	$C_{bid} - P_{bid} - (F_{ask} - K)e^{-rT} - \text{cost}$	$(C_{bid} - P_{bid})S_{bid} - (F_{ask} - K)e^{-rT} - \text{cost}$
bck, join call bid	$P_{bid} - C_{bid} + (F_{bid} - K)e^{-rT} - \text{cost}$	$(P_{bid} - C_{bid})S_{bid} + (F_{bid} - K)e^{-rT} - \text{cost}$
bck, join put ask	$P_{ask} - C_{ask} + (F_{bid} - K)e^{-rT} - \text{cost}$	$(P_{ask} - C_{ask})S_{ask} + (F_{bid} - K)e^{-rT} - \text{cost}$
fwd, join call ask	$C_{ask} - P_{ask} - (F_{ask} - K)e^{-rT} - \text{cost}$	$(C_{ask} - P_{ask})S_{ask} - (F_{ask} - K)e^{-rT} - \text{cost}$

(g, t) :

$$\text{MMA}_{g,t} = \frac{1}{4n_{g,t}} \sum_{i \in g} \sum_{p=1}^4 m_{i,p,t}, \quad (1)$$

where $n_{g,t}$ is the number of pairs in cell g, t . The average includes the many paths with non-positive profit after costs, as well as the positive ones.

We can also compute MMA for t , summing out g , or vice versa. We can also form MMA without the index: a global average over all tickpaths. Just to be precise: MMA is an average over tickpaths, and not over cells. To clear up potential misunderstandings, some integrated formulas for different aggregation levels are explicitly:

$$\begin{aligned} \text{MMA} &= \frac{1}{4n} \sum_{i,p,t} m_{i,p,t}, & \text{MMA}_g &= \frac{1}{4n_g} \sum_{i \in g, p, t} m_{i,p,t}, \\ \text{MMA}_d &= \frac{1}{4n_d} \sum_{t \in d, i, p} m_{i,p,t}, & \text{MMA}_{g,d} &= \frac{1}{4n_{g,d}} \sum_{t \in d, i \in g, p} m_{i,p,t}, \end{aligned} \quad (2)$$

where n , n_g , n_d , are $n_{g,d}$ are the number of ticks, ticks in a cell, ticks in a period d (e.g. a day or a number of days), and ticks in a cell-period, respectively.

Market unfairness (MU) comes in two complementary forms, both built from the positive part of m . The *conditional unfairness* U_c averages m over only the *positive-MMA tickpaths*—those with $m_{i,p,t} > 0$:

$$U_{c,g,d} = \frac{\sum_{i \in g, p, t \in d} \max(0, m_{i,p,t})}{\sum_{i \in g, p, t \in d} \mathbf{1}\{m_{i,p,t} > 0\}}, \quad (3)$$

the average *size* of the arbitrage, given that one exists. Let the *unfairness rate* $R_{g,d}$ be the share of the $4n_{g,d}$ tickpaths with a positive payoff,

$$R_{g,d} = \frac{1}{4n_{g,d}} \sum_{i \in g, p, t \in d} \mathbf{1}\{m_{i,p,t} > 0\}, \quad (4)$$

and let the *unconditional unfairness* U_u average the same positive part over *all* tickpaths:

$$U_{u,g,d} = \frac{1}{4n_{g,d}} \sum_{i \in g,p,t \in d} \max(0, m_{i,p,t}) = R_{g,d} U_{c,g,d}. \quad (5)$$

As is often the case with measuring long-tail averages, U_c and U_u are sensitive to outliers and therefore sensitive to capping or censoring m . Being dollar denominated, they also suffer from scaling issues. As we'll see in the empirical section, R is quite a natural proxy that can be used to characterize MU. Think of market unfairness through an imaginary maker who joins a quote only when the parity payoff is in her favor. It is a imaginary expectation, conditional on getting filled, not a realized profit, which depends on queue position, adverse selection and on a taker crossing. The compensation of the market maker in general is the bid–ask spread which already pays her for the usual costs of making a market—order processing, inventory, adverse selection, queue risk. Market unfairness is separate: it measures only how differently the two equivalent instruments—the real quote and its parity-implied synthetic—are priced against one another.

4.2 Description of arbitrage configurations

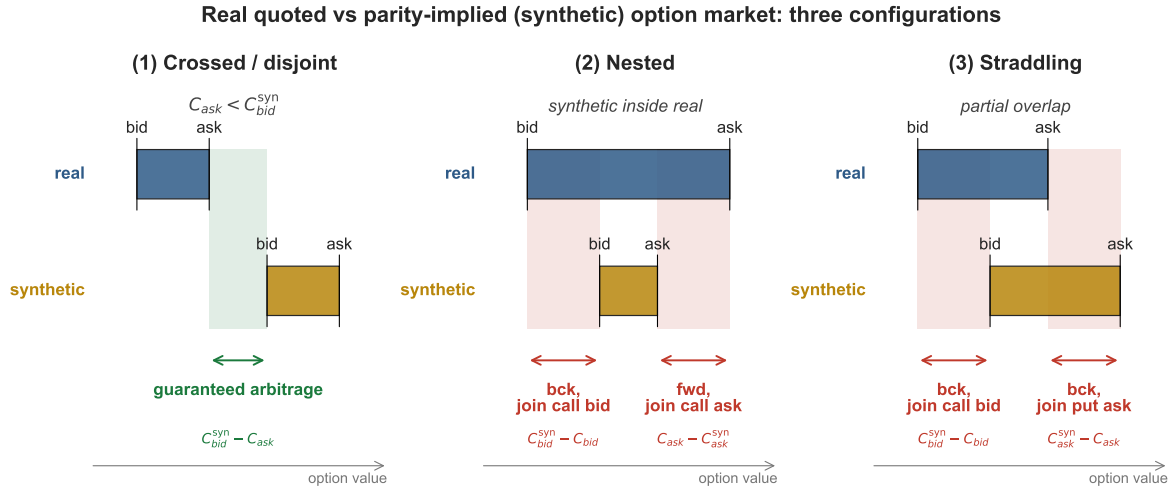


Figure 4: Real quoted versus parity-implied (synthetic) option market. When the markets are crossed and disjoint ($C_{ask} < C_{bid}^{syn}$, or the symmetric case), there is riskless arbitrage in the conventional sense. When one market is nested inside the other, or when the two straddle each other, there is no guaranteed profit but two paths carry a *potential* market-maker advantage, realized only if the maker's quote is joined and then hit. In the nested case these are the backward join-call-bid and forward join-call-ask paths ($C_{bid}^{syn} - C_{bid}$ and $C_{ask} - C_{ask}^{syn}$), with the join-put payoffs equal to their negatives (exactly so when the futures market has zero spread; the trading cost need not be zero); in the straddling case they are the backward join-call-bid and backward join-put-ask paths ($C_{bid}^{syn} - C_{bid}$ and $C_{ask}^{syn} - C_{ask}$). The latter two configurations are what market unfairness aggregates. Schematic, illustrative levels.

It is useful to see what market unfairness is measuring geometrically. For a given option leg—say the call—the quotes define a *real* market $[C_{bid}, C_{ask}]$, while parity combined with the put and the matching futures defines a *synthetic* market $[C_{bid}^{syn}, C_{ask}^{syn}]$, where

$C^{\text{syn}} = P + (F - K)e^{-rT}$. Both intervals live on the same price axis, and their relative position determines whether, and how strongly, an arbitrage exists. Figure 4 shows the three qualitatively distinct configurations.

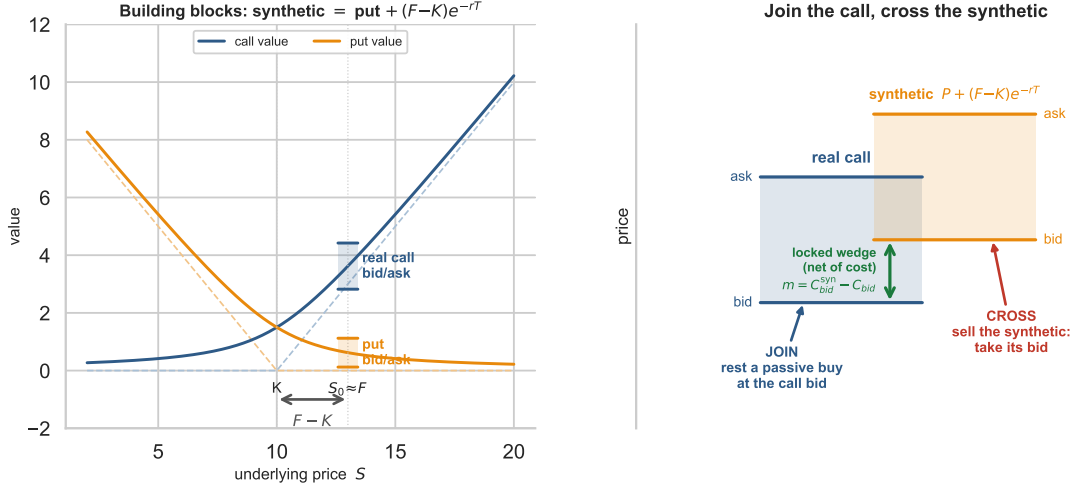


Figure 5: One path in detail: the backward join-call-bid (illustrative, exaggerated). *Left:* the synthetic call is assembled from parity—the call and put quotes and the forward-minus-strike $F - K$, giving $C^{\text{syn}} = P + (F - K)e^{-rT}$; the object we measure is a top-of-book *quote* (a bid/ask price), not a payoff. *Right:* when the real call quote sits below its parity-implied synthetic quote $C^{\text{syn}} = P + (F - K)e^{-rT}$, the maker *joins*—rests a passive buy at the real call bid—and, once that resting order is hit, *crosses* the synthetic market, selling the parity-equivalent synthetic at its bid. The arb is $C_{bid}^{\text{syn}} - C_{bid}$, net of the round-trip cost (the nested backward join-call-bid of Figure 4). Only the resting leg is passive; the arb is therefore a payoff *conditional on the maker being filled*, not a guaranteed trading profit. Schematic, illustrative levels.

MMA and MU do not measure the spread itself, which is the compensation makers demand for liquidity provision. They measure where the real and synthetic quotes are: each real quote must track its synthetic counterpart within the cost cushion (Section 4.3), and MU is what occurs outside that tolerance. This is how similarity with the NBBO rule arises: the synthetic market is the other market, and the market maker should not quote a worse price than what is available elsewhere (i.e in a synthetic way).

4.3 The no-MU band and the shape of the MMA distribution

The configurations in Figure 4 raise a sharper question: for a given synthetic market, how far can a real quote move before it becomes unfair? The answer pins each real quote to its synthetic counterpart within a tolerance set by the trading cost, and it fixes the shape that the MMA distribution “should” take.

For a detailed explanation, consider the real call ask C_{ask} . Two of the four paths in Table 1 contain it—backward/join-put-ask and forward/join-call-ask. Requiring both payoffs to be non-positive (no U_u) yields

$$P_{ask} + (F_{ask} - K)e^{-rT} - \text{cost} \leq C_{ask} \leq P_{ask} + (F_{ask} - K)e^{-rT} + \text{cost}. \quad (6)$$

For later convenience let’s define $\text{cost}' \equiv \text{cost} + \frac{1}{2}(F_{ask} - F_{bid})e^{-rT}$, which is the cost plus half the futures spread. Both bounds use the *same-side* put quote P_{ask} , so the band is

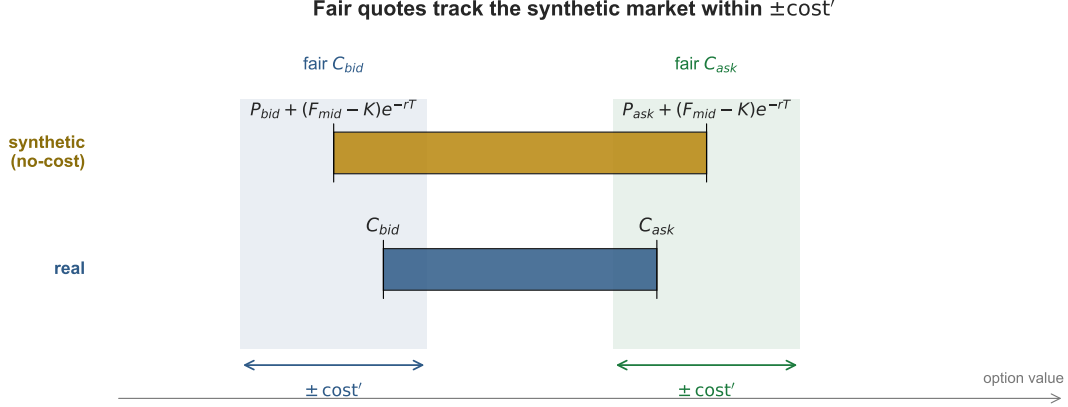


Figure 6: No U_u requires each real quote to track its matching synthetic quote within $\pm \text{cost}' = \pm(\text{cost} + \frac{1}{2}(F_{ask} - F_{bid})e^{-rT})$. The band centers are the frictionless (no-cost) synthetic prices $P_{bid} + (F_{mid} - K)e^{-rT}$ and $P_{ask} + (F_{mid} - K)e^{-rT}$; the cushion around each is $\pm \text{cost}'$. As frictions shrink, the real market is squeezed onto the synthetic market. Schematic, illustrative levels.

centred on the synthetic ask at the futures mid, $C_{ask}^{\text{syn}} \equiv P_{ask} + (F_{mid} - K)e^{-rT}$, with half-width cost' . So a fair real market must track the synthetic market side by side, to within $\pm \text{cost}'$ (Figure 6). It is *not* that C_{ask} must lie inside the synthetic bid–ask up to some cost. Rather, C_{ask} is tied to the synthetic *ask* within the $\pm \text{cost}$ cushion. Limits to the other quote-sides are analogous.

Equation (6) also fixes the shape of the MMA distribution. Write $u = C_{ask} - C_{ask}^{\text{syn}}$; the two conjugate C_{ask} payoffs are exactly $u - \text{cost}'$ (backward, join put ask) and $-u - \text{cost}'$ (forward, join call ask). As C_{ask} ranges over its fair band $u \in (-\text{cost}', \text{cost}')$, each sweeps linearly over $(-2\text{cost}', 0)$. If makers quoted with no preferred point inside the band, the implied MMA would be *uniform* on $(-2\text{cost}', 0)$: a rectangle of width $2\text{cost}'$ just left of zero. That rectangle is the no-information benchmark (dashed in Figure ??).

The observed distribution instead peaks near $-\text{cost}'$. Two things to point here. First just as C_{ask} depends on P_{ask} with a correction, P_{ask} is limited by C_{ask} with the same offset. Each unfair quote has a conjugate pair: if C_{ask} is too rich, then P_{ask} is too cheap by the same amount. Assembling tickpaths by conjugate pairs shows that the distribution of MMA must be symmetric around cost' .

Second, market makers have an interest in not posting negative MMA quotes. Heuristically, if we assume similar pressure on both conjugate paths to be more positive, an equilibrium would form where they are equally negative, and the distribution would peak at $-\text{cost}'$. (Figure ??). Stale or aggressive quotes populate the tails; the right tail crossing zero is the MU.

We'll see empirical evidence of this shape in Section 5.

4.4 Data

We use high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching-expiry futures from Tardis.dev (Tardis.dev 2026), spanning 2020 through June 2026, with synchronized best bids and asks for calls, puts, and futures. Ticks are processed with our open-source codebases—`balazs14/TardisProject` and `balazs14/publications` (subfolder `AMU-liquidity-premium`)—which implement synchronization, filtering, parity-

path construction, and market unfairness aggregation (Králik 2026*b,a*). The same calculation applies across the full horizon, so pre/post-2024 comparisons are coherent.

	BTCUSD		ETHUSD	
	deribit	okex	deribit	okex
Avg num contracts	70	46	42	32
Avg num expirations	6	4	6	4
Avg num strikes	41	34	29	30
Avg call spread (bp of index)	75	54	82	64
Avg put spread (bp of index)	63	52	75	61
Num days in sample	2336	2269	2336	2145

Table 2: Summary statistics of matched put–call pairs in the 2020–2026 sample.

Table 2 summarizes coverage. Deribit lists a wider surface—more contracts, expirations, and strikes per day—while spreads are broadly comparable, slightly wider for ETH on OKX.

4.5 *Panel construction*

Observations are synchronized on a regular time grid, filtered to matched pairs with usable quotes, and aggregated into cells indexed by (day, exchange, ref_sym, rel_strike_bucket, tte_bucket). This preserves the cross-section while supporting the time-series analysis. Table 3 collects the processing assumptions.

Table 3: Data-processing assumptions for MMA/ U_u construction.

Component	Assumption
U_u clipping rule	MMA series are clipped to $[0, 100]$ basis points of the index before positive-part averaging.
MMA clipping rule	MMA series are clipped to $[-500, +500]$ basis points of the index.
Spread filter	Call and put spreads are capped at 1000 basis points of the index.
Relative-strike filter	Allowed range is $[0.79, 1.21]$.
Minimum quote size	The smallest quote size on an arb path must be at least \$5000 equivalent.
Subsampling	Last-known observation carried forward to the configured interval (default <code>5min</code>); a leg with a missing or non-updated bid/ask price or size is marked stale.
Stale-quote filter	disabled: when enabled, ticks with a stale call, put, or spot leg are dropped before any U_u /MMA aggregation.
Discount rate	Risk-free rate assumed $r = 5\%$; it enters only through the factor e^{-rT} . Not a crucial assumption—see the sensitivity analysis (Section 8).
Trading cost	Round-trip cost for one trade unit: <code>cost_per_notional</code> (0.003) $\times$ index, a proportional-to-notional (Section 8).

5 Empirical structure of the MU

5.1 The shape of the MMA distribution

The empirical shape of the MMA distribution depends on the units of measurement. We choose to report MMA as basis points of the index for a one-trading unit arbitrage. This standardizes nicely over the different underlyings, and it is compatible with a cost structure dominated by a fee charged by notional traded. Since fees are customarily given as a number of coins per size traded, the dollar value of the cost is indeed best approximated as a fraction of the index.

Cost structures are not uniform across venues and underlyings, the volume traded by the market maker in question, and the location of the contract on the strike-expiry surface. We choose a single cost parameter for the sake of exposition, and we show in Section 8 that the results are not sensitive to this choice.

Figure 7 shows that the 4 arbitrage paths behave identically. The density peaks, as expected, around minus the trading cost, which was chosen at -30 basis points for this plot.

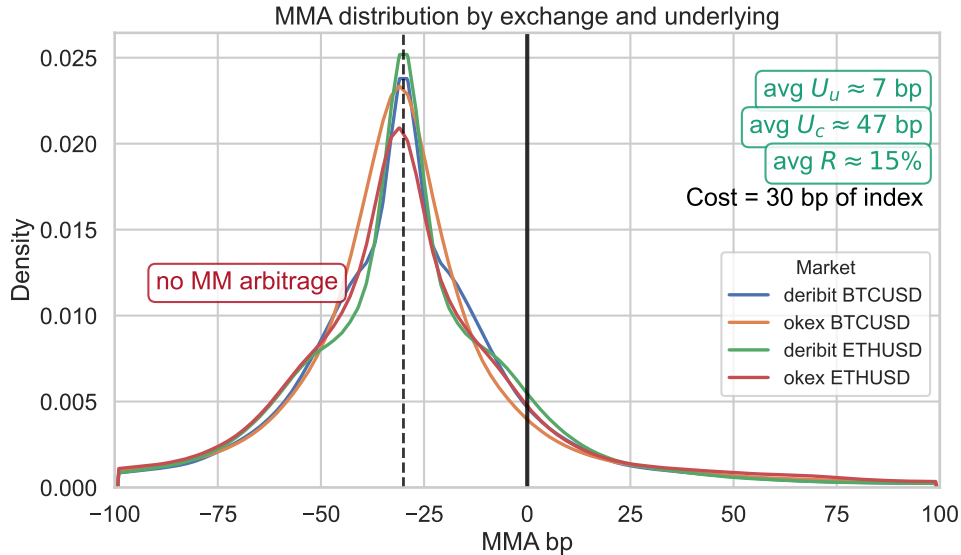


Figure 7: Average MMA density by exchange/underlying market (Deribit BTCUSD, Deribit ETHUSD, OKX BTCUSD, OKX ETHUSD), where each curve averages the four join-path distributions within that market. The four curves are approximately the same because both the cost floor and the histogram variable are expressed in basis points of the index, which puts all markets on a common scale.

5.2 U_u across exchanges, moneyness, and maturity

The global averages (Table 4) show venues and underlyings differ at most moderately. However, over the strike-expiry surface, there are clear patterns of variation. Regarding moneyness, the market is much more unfair in the wings than at the money. Regarding maturity, the market is most unfair at the shortest and longest maturities, and least unfair around one month. The patterns are regular across venues and underlyings.

The unconditional unfairness concentrates at the wings and the tails.

Figures 8 and 9 decompose the unconditional unfairness into its two constitutive margins:

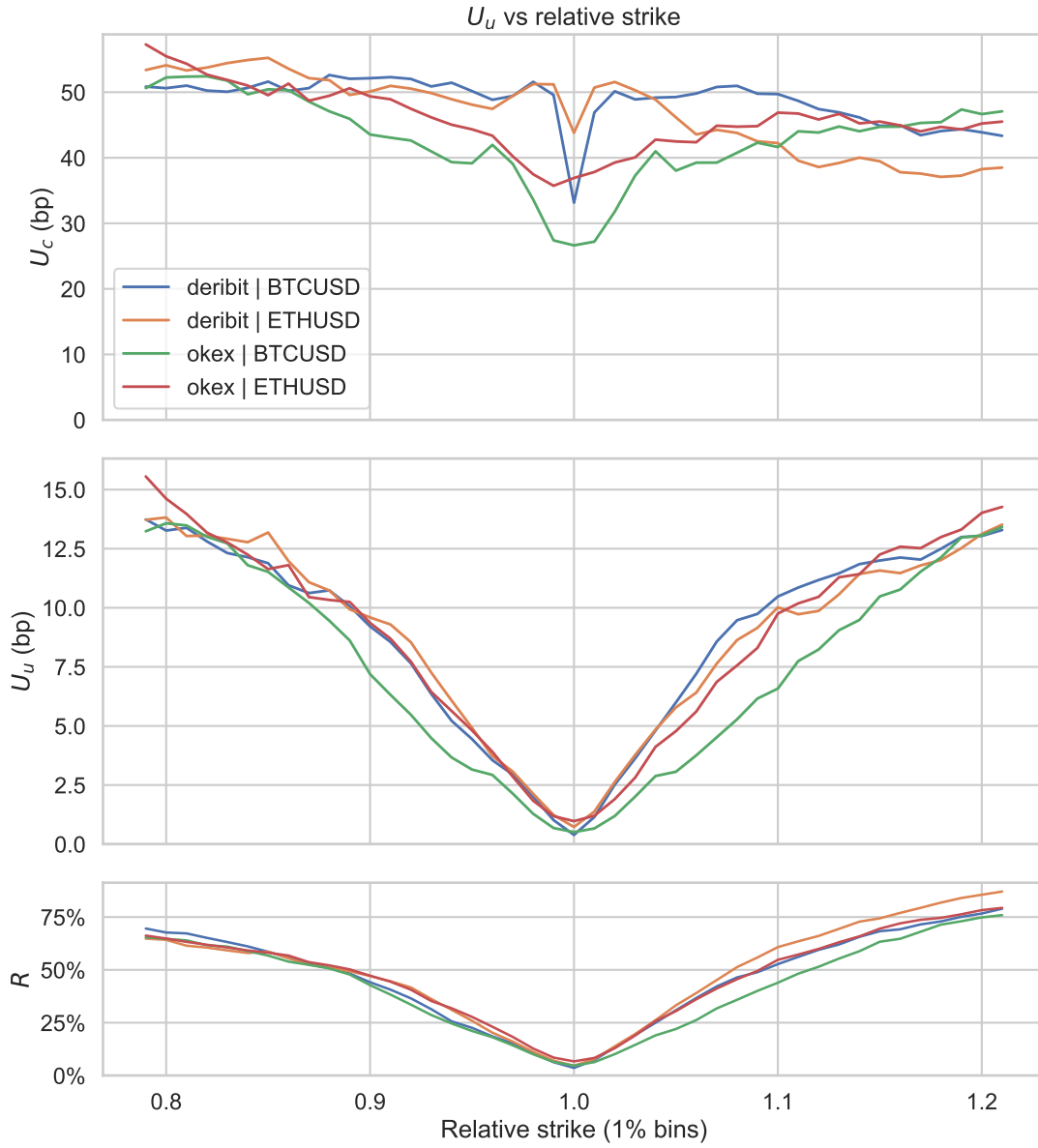


Figure 8: U_u by relative strike across exchanges and underlyings, decomposed into its two margins: the conditional size U_c (upper panel, comparatively flat), the unconditional unfairness $U_u = R U_c$ (middle panel), and the unfairness rate R (four paths per matched quote) (lower panel, a deep V minimized at the money). The unconditional unfairness inherits the V shape from R .

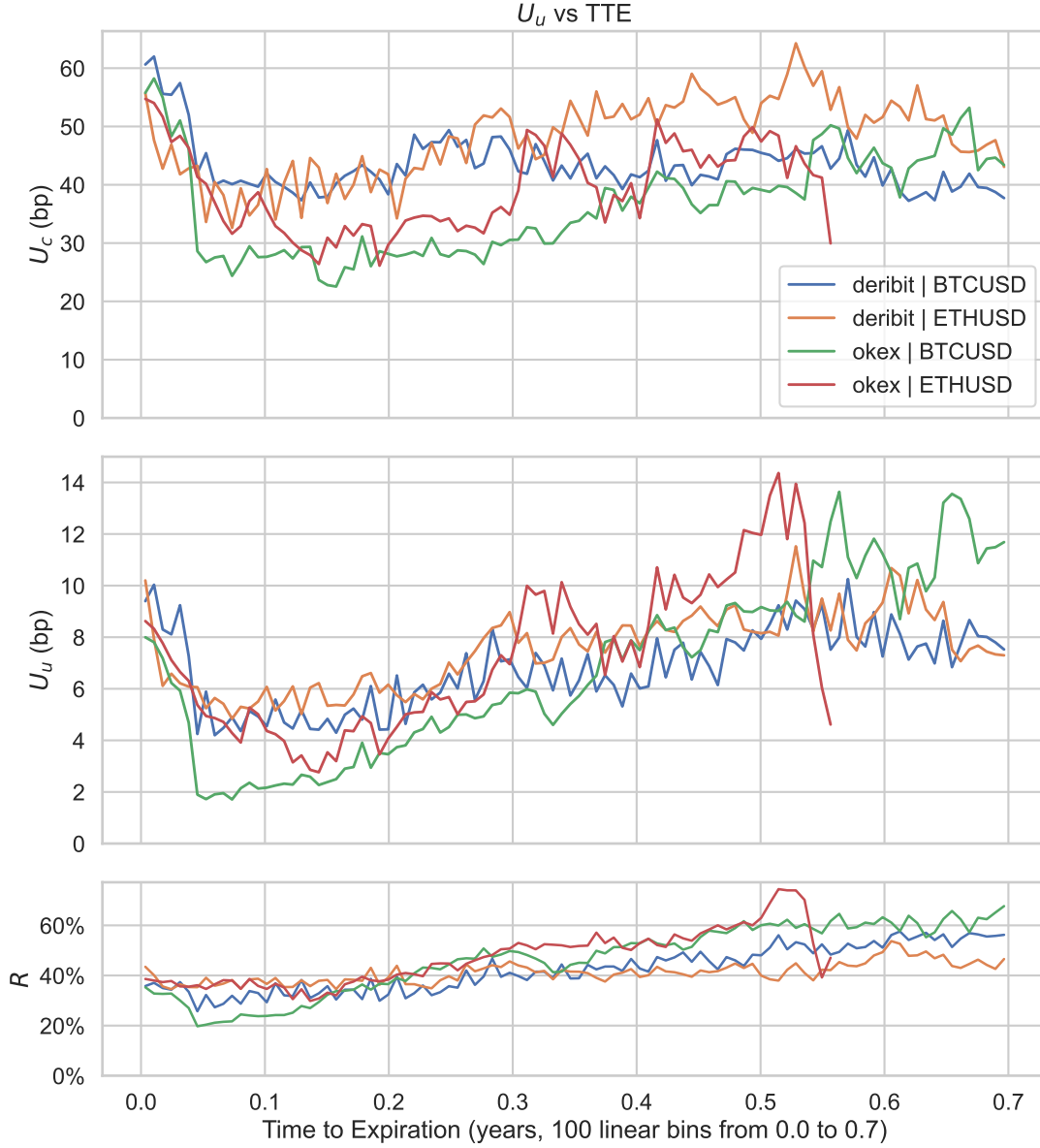


Figure 9: U_u by time to expiry across exchanges and underlyings: the conditional size U_c (upper panel, comparatively flat), the unconditional unfairness $U_u = RU_c$ (middle panel), and the unfairness rate R (four paths per matched quote) (lower panel). The frequency—and hence the unconditional unfairness—is U-shaped: elevated at the shortest and longest maturities, lowest around one month.

		U_c (bp)	U_u (bp)	R	num pairs
underlying	exchange				
BTCUSD	deribit	49.23	7.45	0.151	46,832,305
	okex	44.56	5.80	0.130	30,167,154
ETHUSD	deribit	46.37	7.30	0.157	27,980,129
	okex	46.74	7.29	0.156	19,856,197

Table 4: Global MU summary across exchanges and underlyings, reported as basis points of the index, or as a fraction, for the case of R .

the conditional size U_c and the unfairness rate R . Across the strike dimension, the conditional size remains comparatively flat, but the unfairness rate R exhibits a deep V shape minimized at the money, causing the unconditional unfairness to inherit this V-shaped pattern from the wings. Across the maturity dimension, the conditional size again stays flat, while the unfairness rate is U-shaped with elevated frequency at the shortest and longest maturities and a trough around one month. This U-shaped pattern in the unfairness rate drives the corresponding U-shaped profile of unconditional unfairness by time to expiry. Together, these decompositions reveal that the variation in unfairness is driven primarily by variation in the unfairness rate—reflecting the difficulty of arbitrage and the frequency of parity violations—rather than by variation in the size of individual violations. The same spatial pattern is shown later in Section 6.1.

6 The 2024 regulatory shift

Three milestones cluster in 2024. On January 10 the SEC approved spot Bitcoin ETPs, connecting crypto price risk to regulated custody, authorized-participant arbitrage, and mainstream brokerage (U.S. Securities and Exchange Commission 2024*b*). In May it approved spot Ether ETFs, which began trading in July (U.S. Securities and Exchange Commission 2024*a*). In parallel, MiCA came into force in stages: stablecoin provisions in mid-2024, the CASP licensing regime at end-2024 (European Securities and Markets Authority 2024). None of this changed put–call parity; it changed who can hold, finance, and hedge crypto exposure at scale, and under what legal certainty.

This is where market unfairness is useful as a diagnostic. Whatever keeps a parity wedge open, it is plausibly cheaper to bear for a large, diversified, well-capitalized institution than for the crypto-native firms that dominated before: regulation eases custody and legal risk, balance-sheet depth eases funding risk, diversification eases inventory and convergence risk. If 2024 brought such participants into quoting these markets—directly, or by financing those who do—one would expect MU to shrink, beyond any gain in previously used measures of liquidity.

6.1 *MU over time*

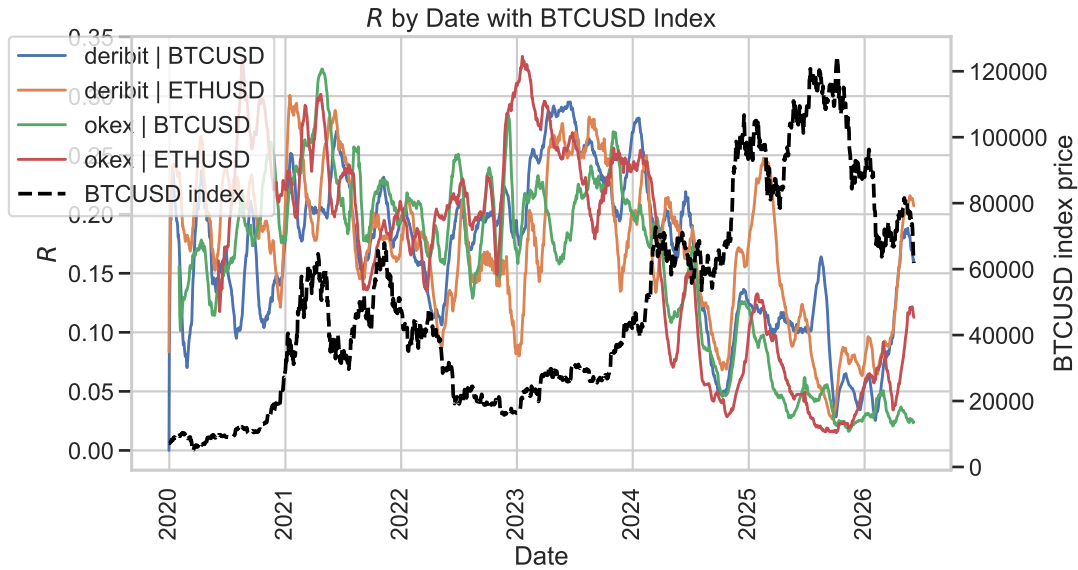


Figure 10: Daily unfairness rate R by exchange and underlying, 2020–2026, with the BTCUSD index overlaid. It is high and volatile before 2024 and compresses to a lower, tighter band afterward.

The drop does not depend on the cost assumption.

The compression is not an artifact of the cost assumption. Figure 12 evaluates the unfairness rate R over a grid of costs, pre and post. The pre-to-post gap remains clearly negative across the full grid.

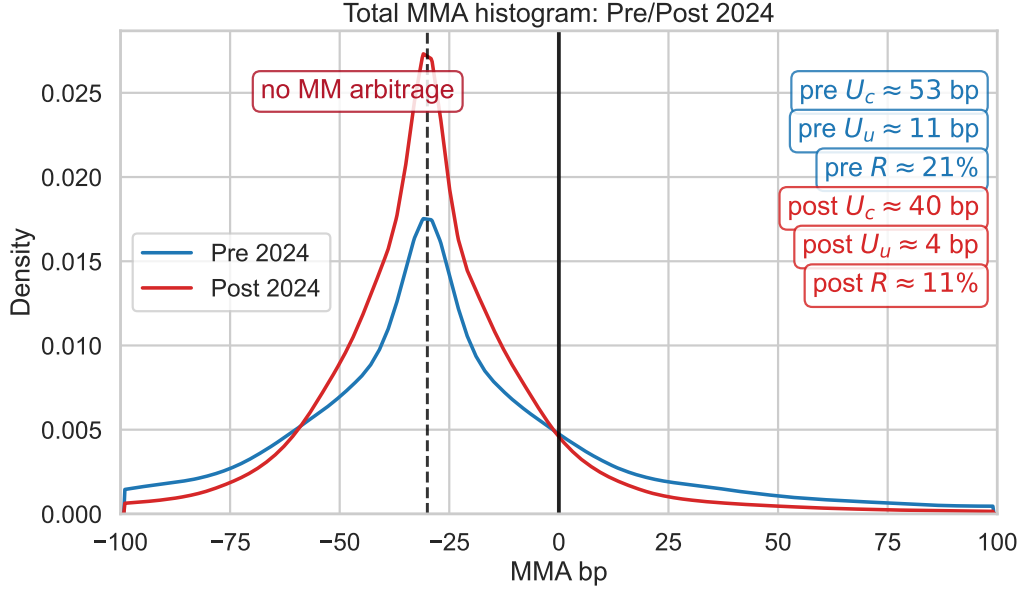


Figure 11: Total MMA histogram across all four paths, split into pre- and post-2024 periods. The post-2024 distribution is more concentrated at the cost floor, and the positive-tail averages—both the conditional and the unconditional unfairness, annotated on the figure—fall markedly after 2024.

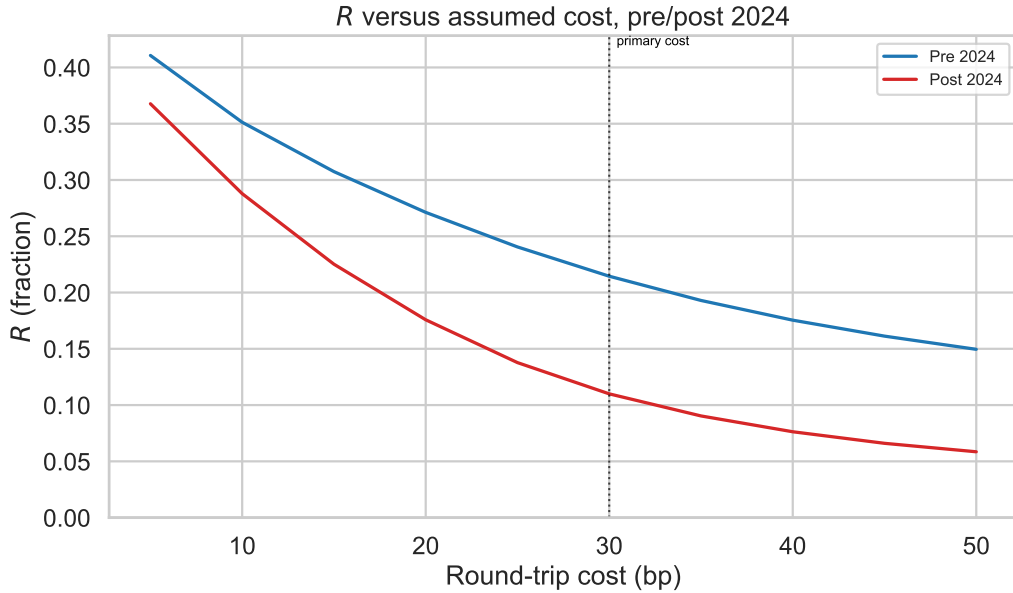


Figure 12: Unfairness rate R versus the assumed round-trip cost, pre- (blue) versus post-2024 (red), computed on the same filtered panel as the regressions; the dotted line marks the primary cost (30 bp). The post-2024 series lies below the pre-2024 series across the whole grid, showing that the frequency of positive parity wedges is lower after 2024 for all tested cost assumptions.

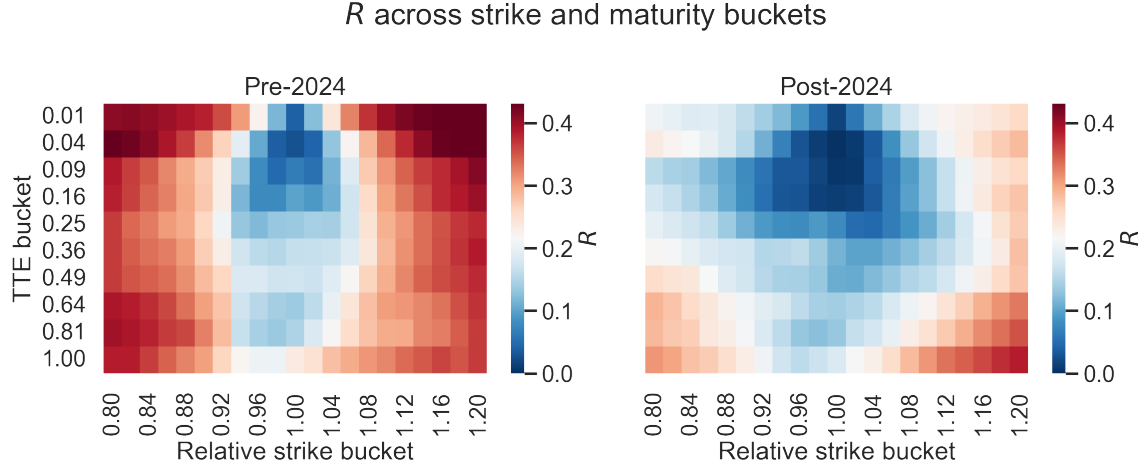


Figure 13: Mean unfairness rate R across relative-strike and maturity buckets, pre- versus post-2024. The high- R interior of the surface cools after 2024; only the deep-OTM, long-dated corner stays elevated.

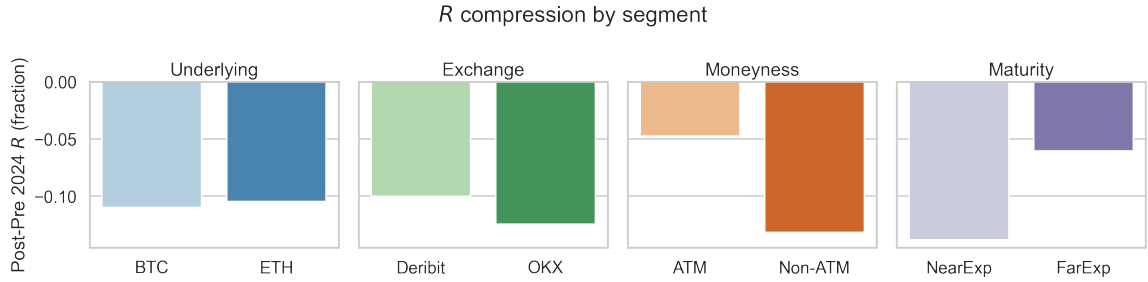


Figure 14: Pre- to post-2024 change in the unfairness rate R by segment (BTC/ETH, Deribit/OKX, ATM/Non-ATM, NearExp/FarExp). Every segment compresses, but the biggest drop is in non-atm near expiry contracts.

7 Regression analysis

7.1 Empirical design

The dependent variable in every specification is $R_{g,t}$, the share of positive cost-adjusted parity wedges in a cell-day. We use cell fixed effects in all specifications, so identification is within fixed (`exchange`, `ref_sym`, `rel_strike_bucket`, `tte_bucket`) cells over time.

We run four fixed-effects specifications.

$$R_{g,t} = \beta_0 \text{Post}_t + \gamma_g + \varepsilon_{g,t}, \quad (7)$$

$$R_{g,t} = \beta_0 \text{Post}_t + \beta_1 \text{Depth}_{g,t} + \beta_2 \text{Spr}_{g,t}^{\text{BTC}} + \beta_3 \text{Spr}_{g,t}^{\text{ETH}} + \beta_4 \text{Stale}_{g,t} + \gamma_g + \varepsilon_{g,t}, \quad (8)$$

$$R_{g,t} = \beta_0 \text{Post}_t + \beta_1 \text{Depth}_{g,t} + \beta_2 \text{Spr}_{g,t}^{\text{BTC}} + \beta_3 \text{Spr}_{g,t}^{\text{ETH}} + \beta_4 \text{Amh}_{g,t}^{\text{BTC}} + \beta_5 \text{Amh}_{g,t}^{\text{ETH}} + \beta_6 \text{Stale}_{g,t} + \beta_7 \text{Rec}_{g,t} + \gamma_g + \varepsilon_{g,t}, \quad (9)$$

$$R_{g,t} = \beta_0 \text{Depth}_{g,t} + \beta_1 \text{Spr}_{g,t}^{\text{BTC}} + \beta_2 \text{Spr}_{g,t}^{\text{ETH}} + \beta_3 \text{Amh}_{g,t}^{\text{BTC}} + \beta_4 \text{Amh}_{g,t}^{\text{ETH}} + \beta_5 \text{Stale}_{g,t} + \beta_6 \text{Rec}_{g,t} + \gamma_g + \varepsilon_{g,t} \quad (10)$$

We use interacted spread and Amihud terms to allow different BTC/ETH slopes, necessitated by the significantly different sizes of one trade unit in the two coins. All models report two-way cluster-robust (by cell and day) standard errors. Table 5 defines the abbreviations.

Table 5: Abbreviations used in the empirical specifications.

Abbreviation	Definition
Post_t	Post-2024 regime indicator (1 for dates on/after 2024-01-10, 0 otherwise).
<i>Liquidity-friction variables</i>	
$\text{Spr}_{g,t}$	Average put–call dollar spread.
$\text{Depth}_{g,t}$	Quote depth in dollars for one trade unit (min over put/call bid/ask quote sizes).
$\text{Stale}_{g,t}$	Mean stale-quote fraction across call, put, and spot legs.
$\text{Amh}_{g,t}$	Amihud illiquidity ratio defined earlier.
$\text{Rec}_{g,t}$	Spread recovery half-life (minutes) defined earlier.
$\text{Spr}_{g,t}^{\text{BTC}}$	Cross-term $\text{Spr}_{g,t} \times \mathbf{1}\{\text{BTC}\}$.
$\text{Spr}_{g,t}^{\text{ETH}}$	Cross-term $\text{Spr}_{g,t} \times \mathbf{1}\{\text{ETH}\}$.
$\text{Amh}_{g,t}^{\text{BTC}}$	Cross-term $\text{Amh}_{g,t} \times \mathbf{1}\{\text{BTC}\}$.
$\text{Amh}_{g,t}^{\text{ETH}}$	Cross-term $\text{Amh}_{g,t} \times \mathbf{1}\{\text{ETH}\}$.
γ_g	Cell fixed effect for the tuple (<code>exchange</code> , <code>ref_sym</code> , <code>rel_strike_bucket</code> , <code>tte_bucket</code>).
$\varepsilon_{g,t}$	Idiosyncratic error.

7.2 Results

Pooled across all cell-days at the baseline 30 bp round-trip cost, the unfairness rate averages $R = 0.148$: a fraction 0.215 of tickpaths violate parity net of cost before 2024, falling to

0.110 after, a change of $\Delta R = -0.105$.

Table 6 is the compact headline regression table. The main conclusion we draw is that MU exhibits an incremental shift around the 2024 regulatory event, even after controlling for the usual liquidity measures. In other words, MU measures an effect (such as the introduction of new participants with different cost structures) that is not captured by the usual liquidity measures.

The liquidity measures we regress against are strongly correlated, so the individual coefficients are not as informative as the R^2 improvement due to the introduction of the Post_t indicator.

Term	(1)	(2)	(3)	(4)
Post_t	-0.127*** [-35%]	-0.100*** [-27%]	-0.102*** [-29%]	
$\text{Depth}_{g,t}$		-0.000*** [-4%]	-0.000*** [-2%]	-0.000*** [-13%]
$\text{Stale}_{g,t}$		0.348*** [20%]	0.507*** [27%]	0.646*** [35%]
$\text{Spr}_{g,t}^{\text{BTC}}$		0.000*** [30%]	0.000*** [30%]	0.000*** [27%]
$\text{Spr}_{g,t}^{\text{ETH}}$		0.003*** [32%]	0.005*** [32%]	0.005*** [35%]
$\text{Rec}_{g,t}$			-0.000	0.000
$\text{Amh}_{g,t}^{\text{BTC}}$			0.728 [1%]	2.202** [2%]
$\text{Amh}_{g,t}^{\text{ETH}}$			0.503*** [1%]	1.168*** [2%]
Total R^2	0.361	0.522	0.596	0.535
Observations	605,006	605,006	343,301	343,301
Fixed effects	cell_id	cell_id	cell_id	cell_id

Table 6: Regressions of the cell-level unfairness rate R . Column (1) regresses R on the post-2024 indicator; (2) adds static liquidity measures, (3) adds dynamic ones and (4) removes the Post term. Each coefficient cell has β with significance stars and, in bold brackets, the standardized effect $\beta \text{std}(X)/\text{std}(Y)$ (the change in R , in standard deviations of R , per one-standard-deviation increase in the regressor).

R tracks the static controls tightly.

Figure 15 is consistent with Table 6: the static controls explain a large share of within-cell variation, while depth contributes with the opposite sign.

R behaves like a market-quality outcome.

Across the four specifications, R is lower after 2024, and the friction controls account for substantial within-cell variation without removing the negative post coefficient. The regressions are descriptive and do not identify participants or mechanisms.

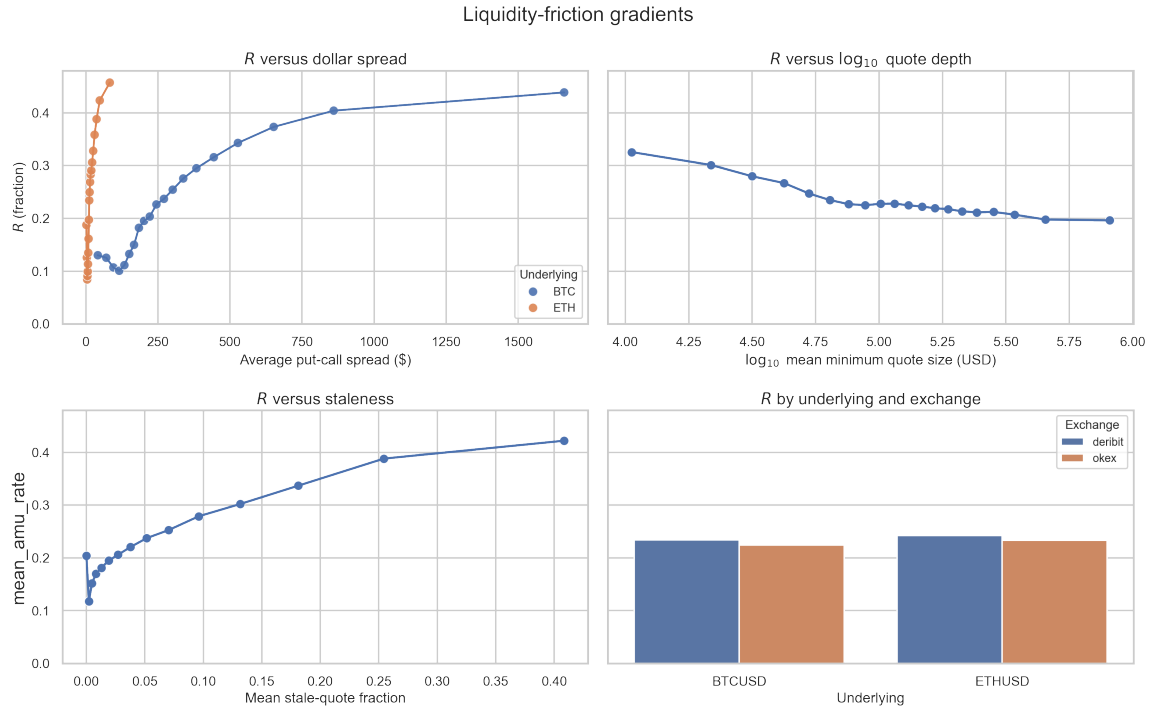


Figure 15: Unfairness rate R against liquidity frictions. Panel 1 plots R against dollar spread with separate BTC and ETH lines. Panel 2 plots R against $\log_{10}$ quote depth (USD). Panel 3 plots R against stale-quote fraction; panel 4 reports average R by underlying and exchange. R rises with spread and staleness, while depth loads negatively in the fixed-effects regressions.

8 Robustness

The regression uses one observation per cell-day, but these are not independent draws. A cell—a fixed strike/maturity/venue bucket—is observed on many days and its wedge is persistent, so its residuals are correlated over time; and all cells share market-wide conditions on a given day, so residuals are correlated across cells within a date. Heteroskedasticity-robust (HC1) errors assume independence across cell-days and are, in consequence, far too small: with hundreds of cells observed over hundreds of days, the effective amount of independent information is closer to the number of cells or days than to the number of cell-days. We therefore report two-way cluster-robust (by cell and day) standard errors, clustering two ways—by cell and by day (Cameron et al. 2011). Only the covariance changes; the coefficients are identical. The mechanism is that the clustered variance sums each cluster’s aggregated score *before* squaring it, so positively correlated within-cluster residuals no longer cancel. In our panel the clustered errors are several times larger than HC1; that is what the reported standard errors and significance stars reflect. The 2024 compression and the friction gradients remain significant under this more conservative variance, but their t -statistics are far smaller than a naive HC1 calculation would suggest.

Spread, depth, staleness, amihud ratio and recovery rate are correlated — a thin book tends to be wide, shallow, and slow to update at once. Collinearity inflates the variances of the individual friction coefficients, but it does not bias the post-2024 coefficient, which is identified from variation orthogonal to the friction set. The significance of the Post_t regressor shows how MU carries information beyond the usual liquidity measures.

Among the exogenous parameters two stand out for robustness checks: the round-trip cost (baseline 30 bp) and the discount rate (baseline $r = 5\%$). Neither turns out to be crucial in identifying MU as carrying more information than the friction terms. The cost only sets the scale of market unfairness—a fixed proportional cost shifts the level, but the pre/post *difference* in the unfairness rate R barely moves with it (the conditional form is more cost-sensitive, one reason the headline uses the rate-based measure), and a common cost keeps cross-venue comparisons coherent. The discount rate matters even less, for a structural reason: r enters only through e^{-rT} , which multiplies a small quantity—the futures leg $(F - K)$ in each payoff and the futures half-spread $\frac{1}{2}(F_{ask} - F_{bid})$ in cost'. Its effect on a payoff is $\partial[(F - K)e^{-rT}]/\partial r = -T(F - K)e^{-rT}$, of order $T(F - K)/\text{index}$; since our strikes sit within $\pm 21\%$ of the forward and most observation mass sits in near-the-money, short-dated cells where $(F - K)$ and T are smallest, plausible changes in r move R negligibly—and the pre/post difference more negligibly still, since r shifts both regimes almost identically. Evaluating R across grids of both parameters—the cost grid read off the same filtered panel as the regressions, the discount-rate grid recomputed at each r (Table 7)—bears this out. A higher assumed cost lowers the R level and shrinks the compression somewhat, but the pre/post drop stays large and negative across the whole cost grid; across r it is essentially constant. Under every choice the 2024 compression is present and sizable.

A final check varies the construction one choice at a time—the positive-part clip, the stale-quote filter, the spread and size cuts, the moneyness window, and tick- versus notional-weighting—recomputing the pre/post unfairness rate R for each (Table 8). The level moves with the filters, as it must, but the post-2024 drop is large and negative under every setting; none of the choices manufactures the compression or erases it.

cost (bp)	R pre	R post	ΔR		r	R pre	R post	ΔR
10	0.351	0.288	-0.063		0.000	0.219	0.134	-0.085
20	0.271	0.176	-0.095		0.025	0.216	0.120	-0.096
30	0.215	0.110	-0.105		0.050	0.215	0.110	-0.105
40	0.175	0.076	-0.099		0.075	0.214	0.109	-0.105
50	0.150	0.058	-0.091		0.100	0.216	0.117	-0.099

Table 7: R pre- and post-2024, as a fraction, across a grid of round-trip costs (baseline 30 bp) and of discount rates r (baseline 5%). Δ is the post–pre change.

setting	R pre	R post	ΔR
baseline	0.215	0.110	-0.105
clip 50 bp	0.215	0.110	-0.105
clip 200 bp	0.215	0.110	-0.105
no clip	0.215	0.110	-0.105
drop stale quotes	0.185	0.104	-0.081
spread cap 500 bp	0.207	0.106	-0.101
spread cap 200 bp	0.159	0.090	-0.070
min size \$25k	0.208	0.110	-0.098
moneyiness $\pm 10\%$	0.138	0.066	-0.072
notional-weighted	0.201	0.107	-0.093

Table 8: Robustness of the pre/post-2024 unfairness rate R to the filter and aggregation choices, each varied one at a time from the baseline (top row): the positive-part clip, the stale-quote filter, the spread and minimum-size cuts, the moneyiness window, and whether we average R with weights corresponding to the maximum tradable size on a tickpath, or whether each tickpath has equal weights. The table shows the stability of Δ , the post–pre R change.

9 Conclusion

We propose *Market Unfairness*: the cost-adjusted put–call-parity violation left in the standing quotes as a measure of market imperfection. We propose the *rate* of these violations as a stable unit-free number, which carries more information than the usual liquidity frictions. It is stable across venues and underlyings, but changes in a statistically significant manner across a known regime shift.

Since MU has an ideal value (0) which is attainable under realistic market dynamics, it is a useful way to judge option trading venues for purposes of market quality, unlike others such as average spread, for which there is no agreed-upon *right* value. Retail investors and regulators may use it to rank venues and products. MU also shows which parts of the option surface are least served by market makers and by how much.

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